Addie McCurdy

University of Colorado Boulder
Department of Applied Mathematics

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Education

Ph.D. in Applied Mathematics **Expected Summer 2027**
University of Colorado Boulder GPA: 3.974
Advisor: Professor Emily Jensen
Thesis title: "Spatial properties of optimal controllers over Sobolev spaces"

B.A.s in Pure Mathematics and Economics **May 2022**
University of St. Thomas - Minnesota GPA: 3.99
Honors: Summa Cum Laude, Aquinas Honors Scholar

Research Experience

Graduate Research Assistant (May 2024 – present)
University of Colorado Boulder
Mentor: Professor Emily Jensen
Focus: scalable optimal control for systems governed by partial differential equations

Graduate Research Assistant (Spring 2024)
University of Colorado Boulder
Mentor: Professor François Meyer
Focus: machine learning methods on network valued random variables

Research in Industrial Projects for Students (RIPS) (Summer 2021)
University of California, Los Angeles
Mentors: Dr. Robert Martin and Dr. Daniel Eckhardt - Air Force Research Laboratory
Focus: deconvolution of dynamically coupled noisy signals, with application to Hall-effect thruster data

Undergraduate Research Assistant (Feb 2019 – May 2022)
University of St. Thomas - Minnesota
Mentor: Professor Eric Rawdon
Focus: analysis and visualization of data for projects concerning new models and biological applications of mathematical knots

Honors and Awards

Sheryl R. Young Memorial Fellowship (2024 – 2025)

Research Visits

University of Michigan (Sep 2026)
Hosted by Professor Peter Seiler, Department of Electrical Engineering and Computer Science
Focus: integral quadratic constraints for robust control of systems governed by partial differential equations

[1] Updated November 12, 2026

Publications and Preprints

Peer-reviewed conference publications:

A3. **Conditions for Complete Decentralization of the Linear Quadratic Regulator**
(**A. McCurdy**, I. Collins, E. Jensen). *27th International Symposium on Mathematical Theory of Networks and Systems*. pp. 139-144, 2026

A2. **A System Level Approach to LQR Control of the Diffusion Equation**
(**A. McCurdy**, A. Gusty, E. Jensen). *American Control Conference* . pp. 4723-4730, 2026

A1. **Complete Decentralization of Linear Quadratic Gaussian Control for the Discrete Wave Equation**
(**A. McCurdy**, E. Jensen). *IEEE 64th Conference on Decision and Control*. pp. 6162-6168, 2025

Under review:

B1. **Complete Decentralization of Wave Equation Control on a Compact Domain**
(**A. McCurdy**, E. Jensen). *under review at SIAM Journal on Control and Optimization*. arXiv:2611.13449

Talks and Presentations

C7. *Conditions for Complete Decentralization of the Linear Quadratic Regulator*. (poster). SIAM Student Mini-Symposium in Applied Mathematics, Ann Arbor, Michigan, August 2026.

C6. *Conditions for Complete Decentralization of the Linear Quadratic Regulator*, 27th International Symposium on Mathematical Theory of Networks and Systems, Waterloo, Canada, August 2026.

C5. *A System Level Approach to LQR Control of the Diffusion Equation*, American Control Conference, New Orleans, Louisiana, May 2026.

C4. *Complete Decentralization of Linear Quadratic Gaussian Control for the Discrete Wave Equation*, IEEE 64th Conference on Decision and Control, Rio de Janeiro, Brazil, December 2025.

C3. *An introduction to optimal control of systems governed by PDEs*, CU Boulder SIAM graduate student chapter seminar, Boulder, Colorado, October 2025.

C2. *Deconvolution of Temporally Under-Resolved Image Sequences for Coupled Dynamical Systems*, Joint Mathematics Meeting, (virtual), January 2022.

C1. *Exploring the garden of Petaluma knots*, Nebraska Conference for Undergraduate Women in Mathematics, Lincoln, Nebraska, January 2020.

Teaching and Mentorship Experience

Discovery Learning Apprenticeship (DLA) Mentor (Aug 2026 – May 2027)
University of Colorado Boulder
Primary mentor to undergraduate student Elena Abbruzzi in university funded research.

Mentor for Junior Researchers (Sep 2025 – Aug 2026)
University of Colorado Boulder
Mentored undergraduate researcher Andrew Gusty resulting in publication (A2), and first year graduate student Isabel Collins, resulting in publication (A3).

Graduate Teaching Assistant (Aug 2022 – Dec 2023)
University of Colorado Boulder

Taught recitation and workgroup sections, held office hours, and wrote and graded quizzes, homeworks, and projects for courses such as Calculus 1 and 2 for Engineers and Introduction to Differential Equations with Linear Algebra.

Math Resource Center (MaRC) Tutor (Feb 2020 – May 2022)
University of St. Thomas - Minnesota
Peer tutor for all math courses through multivariable calculus. Included training in diversity, equity, and inclusion and tutoring best practices

Service

Association for Women in Mathematics
President: Boulder AWM student chapter (May 2024 – Apr 2026)
Treasurer: Boulder AWM student chapter (May 2023 – Apr 2024)
Member: National AWM organization and Boulder AWM student chapter (Aug 2022 – present)

Reviewer
IEEE Conference on Decision and Control

Volunteer
American Control Conference - Denver, Colorado (July 2025)